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REVIEW ARTICLE

A systematic review on cigarette butt management as a hazardous waste and prevalent litter: control and recycling

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Abstract

Annually, over 5.5 trillion cigarettes are produced worldwide, and it is estimated that 4.5 trillion cigarette butts (CBs) are still being littered in the world. The dispersal of the CBs has caused this hazardous waste to be considered as one of the most important litters and environmental risks all over the world. This systematic study with the search protocol definition and keyword identification was developed to find the CBs control and recycling methods by searching in five scientific databases. Founded articles were monitored and finally, 35 related articles were selected and studied by the authors. The results of this study showed that CBs recycling methods have been tested in 10 specific categories all over the world. The CBs have been applied without any pre-processing methods in the bricks and asphalt production. However, other applications of the CBs such as the absorbent material production, vector control, and use as a biofilm carrier in wastewater treatment need various processing methods. The researchers also presented models and suggestions such as taxes, penalties, and public education for the control of CBs littering. Despite the innovative methods applied for the CBs recycling in previous studies, CBs have not received much attention in terms of pollutant control and environmental issues in recycling processes.

Keywords Cigarette butt · Cigarette filter · Hazardous waste · Litter · Waste management · Recycling

Introduction

In the mid-twentieth century, cigarette filter was designed to trap the cigarette smoke chemicals and toxic substances, so that, currently the dominant form of cigarette in the world is filtered cigarette (Novotny et al. 2009; Mohajerani et al. 2017). Annually, over 5.5 trillion filtered cigarettes are produced (Novotny et al. 2009; Barnes 2011; Kadir and Mohajerani 2011; Aeslina and Mohajerani 2012; Yazdi et al. 2012; Dieng et al. 2014; Mohajerani et al. 2016; Ou et al. 2016; Yang et al. 2017) and consumed across the world, leading to the production of an inevitable waste that is called "cigarette butt"(CB) and mostly littered by smokers (Barnes 2011) (Mohajerani et al. 2017) (Mohajerani et al. 2016). Considering the CB-related contaminants such as heavy metals, polycyclic aromatic hydrocarbons (PAHs), formaldehyde, acetaldehydes, hydrogen cyanide, and benzene (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Dieng et al. 2013; Mohajerani et al. 2016), as well as its high dispersion in the world, CBs are one of the most important wastes in the world (Ariza and Leatherman 2012; Dieng et al. 2014; Mohajerani et al. 2017; Maderuelo-Sanz et al. 2018; Murugan et al. 2018) and it is considered as a serious environmental concern due to toxic effects on living organisms (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Ariza and Leatherman 2012; Kadir et al. 2014; Mohajerani et al. 2016, 2017). The cigarette filter as the main part of CBs is made up of cellulose acetate, which is a photodegradable (Sabzali et al. 2012; Dieng et al. 2013); it is considered as non-biodegradable material (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Ariza and Leatherman 2012; Dieng et al. 2013; Kadir et al. 2014; Mohajerani et al. 2016). This feature leads to CBs' durability in natural environments (Kadir and

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Mohajerani 2011; Aeslina and Mohajerani 2012; Sabzali et al. 2012; Dieng et al. 2013; Kadir et al. 2014), so that it leaks toxic and chemical substances to the environment. In addition to the toxicity, CB contaminants are likely to enter the food chain (Barnes 2011; Sarani and Kadir 2013; Betânia et al. 2016; Mohajerani et al. 2016; Ou et al. 2016). Due to the difficulty and the high cost of CB collection (Kadir and Mohajerani 2011; Yazdi et al. 2012; Sarani and Kadir 2013; Kadir et al. 2014; Mohajerani et al. 2016) and the lack of a simple method to extract chemical and toxic substances existing in the CB (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Sarani and Kadir 2013), finding solutions to control and reduce the littered CBs seems essential. Furthermore, given that CB landfilling is not recommended economically and environmentally (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Kadir et al. 2014), and hazardous wastes incineration emission is of great concern due to the toxic pollutants emission in the exhaust gases (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Kadir et al. 2014), it is necessary to identify new methods for CBs recycling. Therefore, due to the high dispersion of CBs as a litter in the world, high CB toxicity as well as considering the complex conditions for the final disposal of this hazardous waste, the present study systematically reviews the researchers' experiences on CBs recycling and control pathways.

Method

Literature search

This systematic study was carried out following the definition of a search protocol, identification of the valuable keywords and sources of available information. This review was performed by searching documents in five major scientific databases which focused on CBs as most important solid wastes. The search for papers was carried out in June 2017. The research results and strategy have been presented in Table 1.

Screening and selection criteria

The documents adopted from different databases were selected after removing duplicates and then screened considering the scope of the present study. The methodology to screen most relative documents is descried follows:

(I). Title and abstract: papers with a title and/or abstract which were not consistent with CBs were excluded, even if the selected keyword were found in these studies.
(II). Abstract: each paper abstract was read once more to further select papers that addressed CBs.
(III). Content: the full paper was read to identify the studies that cover CB recycling control methods.
(IV). Related documents: relevant documents referenced on the initially selected papers were searched and read. Next, all studies on CBs including information on one or more of the following topics were selected for the present study:

- _ CB control methods
- _ CB recycling methods

Finally, considering these characteristics, this study was conducted on 35 papers from five databases (Fig. 1).

Literature review

The quantity and importance of CBs

Many studies have reported that, annually, the production of filtered cigarette exceeds 5.5 trillion tons (Novotny et al. 2009; Barnes 2011; Kadir and Mohajerani 2011; Aeslina and Mohajerani 2012; Yazdi et al. 2012; Dieng et al. 2014; Mohajerani et al. 2016, 2017; Ou et al. 2016; Yang et al. 2017); it leads to the production of large amounts of CBs which accordingly endanger both the human body and environment. The estimated amount show that annually 0.8–1.2 millions of CBs is generated in the world (Novotny et al. 2009; Aeslina and Mohajerani 2012). It is also estimated that by 2025, CB

Table 1 Search protocol and the number of articles found on the databases

Search protocol	Source	Article found
(I) Study on the CBs as a waste, (II) time limit for search was not considered, (III) the keywords "CB" OR "cigarette filter" OR "cigarette waste" OR "cigarette litter" OR "cigarette debris" had to appear in the title-abstract-keyword, (IV) the keywords "solid waste" OR "municipal solid waste" OR waste OR litter OR debris OR MSW OR MSWM had to appear in the title-abstract-keyword	Scopus	112
	scienceDirect	35
	PubMed	29
	Web of Science	38
	ProQuest	58

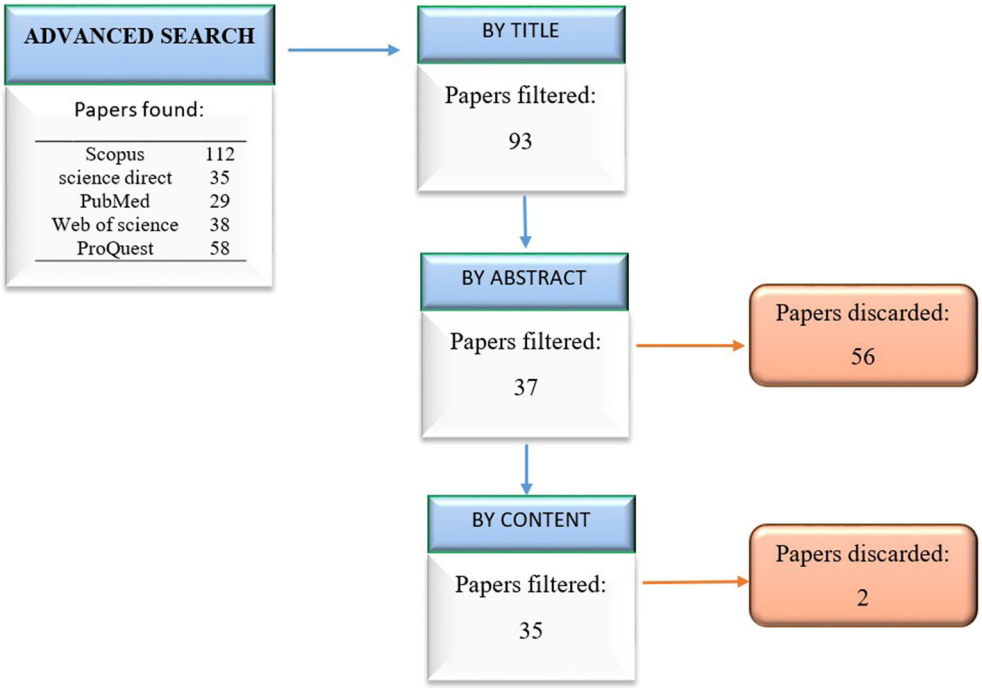


Fig. 1 Outlines of the screening process to finally select the 33 papers used in this study

generation will increase by 50% as a consequence of world population growth (Novotny et al. 2009; Kadir and Mohajerani 2011). Most of CBs are still being littered at a million locations (Barnes 2011; Mohajerani et al. 2016, 2017). CB littering is the most common method of disposal and it is estimated that littered CBs reaches 4.5 billion annually (Yazdi et al. 2012; Dieng et al. 2014; Ou et al. 2016; Mohajerani et al. 2017). Over 95% of the cigarette filters are produced from cellulose acetate, a flax-like substance that is not biodegradable (Kadir and Mohajerani 2011; Aeslina and Mohajerani 2012; Ariza and Leatherman 2012; Yazdi et al. 2012; Dieng et al. 2013; Yang et al. 2017; Maderuelo-Sanz et al. 2018). These filters are made up of microscopic size fibers in the shape of Y (Yang et al. 2017). Thus, CBs have a slow degradability (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Ariza and Leatherman 2012; Sabzali et al. 2012; Dieng et al. 2013; Sarani and Kadir 2013; Mohajerani et al. 2016); however, this material is photodegradable (Sabzali et al. 2012; Dieng et al. 2013) and it is estimated that their durability lasts for up to 18 months and more (Kadir and Mohajerani 2011; Sabzali et al. 2011; Aeslina and Mohajerani 2012; Dieng et al. 2013; Kadir et al. 2014). CBs are designed to trap dangerous substances such as tar and toxins (Mohajerani et al. 2017), and, due to the poor degradability of these wastes, CBs are considered an environmental hazard, which can negatively affect the environment and human body (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Ariza and Leatherman 2012; Kadir et al. 2014; Mohajerani et al. 2016, 2017). Typically, some components such as nicotine are found in tobacco leaves. Additionally, CBs have a range of chemicals which are adapted from different stages of the tobacco plant growth. On top of that, additives in cigarette production considerably raise the toxicity (Dieng et al. 2014; Mondal et al. 2015). Cigarette smoke has 4000 chemical compounds, of which 75% are in the gaseous phase and 25% in the tar phase; some compounds including hydrogen cyanide (HCN), carbon monoxide (CO), arsenic (As), chromium (Cr), peridrin, polycyclic aromatic hydrocarbons (PAHs), N-nitrosamines, aromatic amines, formaldehyde, and acetaldehyde, benzene, and toxic metals such as cadmium and nickel taken together form more than 60 chemicals which have carcinogenic effects and they can be trapped by the smoking in the cigarette filter (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Dieng et al. 2013; Mohajerani et al. 2016).

CB recycling methods

Use of CBs in brick production

Recycling toxic wastes through integrating them into an inert component so as to create the building materials can be considered as an applicable solution (Aeslina and Mohajerani 2012). Thus, CBs can be used as an additive and raw material in brick production (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Kadir et al. 2014; Mohajerani et al. 2016). Moreover, CBs in bricks can abate the leakage of some pollutants, such as heavy metals (Kadir and Mohajerani 2011; Aeslina and Mohajerani 2012), which is due to the trapping characteristics of CBs. In addition, the high-temperature combustion process of bricks production (1050 °C) is an effective way for metal leakage reduction due to the formation of metal oxide (Aeslina and Mohajerani 2012). Of course, the presence

of cellulose acetate, paper, and remained tobacco in CBs lead to increased pollutant emission within the heating process; it was reported that with increase of the CBs in bricks material, the emission of CO, CO_2, and HCN are accordingly increased (Mohajerani et al. 2016). CB is thermal value waste (Abdul Kadir and Mohajerani 2012; Kadir et al. 2014), so that it leads to energy saving in the brick production (Mohajerani et al. 2016). With increasing the ratio of CBs in bricks, the density of bricks decreased significantly; for example 10% of CBs in bricks lead to 636 kg/m^3 (30%) reduction in density (Kadir and Mohajerani 2011). Also, CB recycling in the bricks production can increase the porosity due to cellulose acetate burning and decrease the density and heat conductivity (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Sarani and Kadir 2013; Mohajerani et al. 2016). This decreases in density and increases in porosity lead to the creation of lighter bricks that reduce transmission and installation costs (Kadir and Mohajerani 2011). However, this weight loss and increased porosity can lead to a decrease in the final product strength (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Kadir et al. 2014). In addition, CBs in bricks increase the water absorption capacity and subsequently the shrinkage is increased with the increase in CB ratio. The thermal conductivity of bricks depends on their density and porosity, and by adding 5 and 10% CB in bricks, their thermal conductivity was reduced by 51 and 58%, respectively, which can be considered as an effective way in energy saving issues (Kadir and Mohajerani 2011). The mixing rate or the heating rate in the brick production process with different CB ratios influences the final properties (Kadir and Mohajerani 2011; Aeslina and Mohajerani 2012; Kadir et al. 2014; Mohajerani et al. 2016). The heating rate is an effective factor in brick production; in lesser heating rate, bricks with better characteristics are produced. Researchers found that 2.5% CBs by weight in the brick and the 1 °C/min heating rate are ideal conditions for brick production (Kadir et al. 2014).

Use of CBs in asphalt production

A method for CB recycling is applying it in the asphalt production (Mohajerani et al. 2017). In this sense, Mohajerani et al. (2017) encapsulated CBs in bitumen and then used it to make asphalt for the pavement. CBs were placed at 105 °C for 24 h in order to remove the trapped moisture. Then, CBs were encapsulated in 150 °C hot pot. The study showed that when the amount of CBs is increased, the strength of the produced asphalt was reduced, so that this characteristic should be considered in asphalt standards for light to heavy traffic types. Furthermore, the increased encapsulated CB ratio in asphalt decreases its bulk density which can be different in terms of bitumen class. Eventually, the researchers integrated 10 to 15 kg CB per cubic meter of concrete to achieve acceptable properties in the asphalt production (Mohajerani et al. 2017).

Use of CBs for vector control

Various chemicals found in CBs can be used as a vector control. Fortunately, CBs due to toxic components take advantages of insecticidal properties. (Dieng et al. 2011, 2013; Murugan et al. 2018). It is evidenced that CBs with thousands of chemicals may be highly toxic and have toxic effects on insects (Dieng et al. 2014). Mondal et al. (2015) reported that chemicals and heavy metals existing in CBs have direct and indirect effects on mosquito larvae; these compounds can eliminate the larvae (Mondal et al. 2015). An important reason for the use of CBs as a vector control is due to the negative effect of nicotine and its derivatives on the insects (Dieng et al. 2014). Therefore, researchers have studied CB application in vector control, as shown in Table 2.

It is evidenced that CBs have negative effects on the tested insects, and this effect was observed on the mosquitoes in the egg decomposition (Dieng et al. 2014) and in the larval stage (Dieng et al. 2013, 2014; Mondal et al. 2015; Murugan et al. 2018). However, with increasing concentrations of CB, more effects were observed (Dieng et al. 2013, 2014; Mondal et al. 2015) and the difference in toxicity observed in various studies was attributed to the difference in the brand of cigarette (Dieng et al. 2013). Murugan et al. (2018) showed that although the extracted chemicals from CB have toxic effects on larval and pupal stages in insects, the toxicity of silver nanostructures derived from CB is much higher (see Table 3). In addition, as shown in Table 4, the toxic effects of silver nanostructures produced from CB were reported more severe than those extracted from CB in egg hatchability (Murugan et al. 2018). In addition, this study showed that silver nanostructures inhibited the growth of *Bacillus subtilis*, *Klebsiella pneumoniae*, and *Salmonella typhi* at low concentrations (Murugan et al. 2018). It seems that the toxicity of CBs to insects is true for some birds; in a survey, Su et al. (2017) found that a bird uses CBs in its nest; when the number of live parasites in the nest increased, these birds increased the concentration of CBs in their nests (Su et al. 2017). However, further research is needed to ensure that the effect of this substance has no effect on non-target organisms (Dieng et al. 2014).

Use of CBs in activated carbon production

The littered CBs have the potential to be used as a raw material for the activated carbon preparation (Yang et al. 2017); these materials are chemically or physically converted to activated carbon (Yazdi et al. 2012; Yang et al. 2017). Activated carbons produced from CBs can be modified with the use of specific chemicals such as KOH and HNO_3 (Masoudi Soltani et al. 2015; Yang et al. 2017), which can improve the carbon sorption capacity through increase of the number of surfactant groups (Masoudi Soltani et al. 2015). Therefore, the activated carbon derived from CB can be used to remove pollutants.

Table 2 Use of CBs with the aim of vector control

Vector	Processing	Result	Reference
Aedes aegypti	Soak CBs in 100 ml distilled water for 24 h and use the chemicals leached from them	The insect during the larval period is sensitive to the presence of CBs in its habitat	(Dieng et al. 2013)
Aedes albopictus	Soak CBs in tap water at times of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 days	The effect of CBs in the early larval period is greater than the late larval period	(Dieng et al. 2011)
Aedes aegypti	Soak CBs in 100 ml distilled water for 24 h and use the chemicals leached from them	The study showed that CB affects the decomposing of the eggs	(Dieng et al. 2014)
Culex quinquefasciatus	Soak 5 g of CBs in 250 ml of distilled water for 24 h, then centrifuge the solution and use its supernatant	CBs have direct and indirect effects on mosquito larvae	(Mondal et al. 2015)
Anopheles stephensi	Use extracted substances from CBs for the construction of nano silver structures in the presence of the $AgNO_3$ solution	Substances extracted from CB have toxic effects for the larval stage and pupal stage	(Murugan et al. 2018)

Factors such as pollutant concentrations and pH can affect the absorption capacity (Masoudi Soltani et al. 2015). Various qualitative characteristics of carbon derived from CBs were reported by many researches (Yazdi et al. 2012; Masoudi Soltani et al. 2015; Yang et al. 2017) (see Table 5).

The carbon production process can reduce CBs weight, as reported by the Yang study, 2017; a part of the cigarette mass (16%) is lost during the activated carbon production process (Yang et al. 2017). In addition, the modifying process changes carbon quality, for example, it was shown if charred carbon nanotubes were modified with HNO_3, its surface area decreases to 158 m^2/g and the level of micropores is reduced to 32.4%. What is more, it was reported that the modifying process reduced mesoporous levels from 144 to 133 m^2/g, but the mean pore diameter did not change significantly (Masoudi Soltani et al. 2015).

Use of CBs as biofilm carrier

Several studies on filter rods in cigarette filter have showed that they could be used as biofilm carriers in wastewater treatment (Sabzali et al. 2011, 2012). The researchers stated that, given the roughness and the porous structure of the filter rods which provide suitable conditions for the microorganisms attaching and growth, CBs could act as an attached growth bed (Sabzali et al. 2011, 2012). It was reported that CBs not only has the same performance as commercially available carrier, but also it is cheaper (Sabzali et al. 2012). The biofilm carrier derived from the cigarette filter was used in two types of integrated fixed film activated sludge (IFAS) and anaerobic moving bed biofilm reactors (AMBBRs) (Sabzali et al. 2011, 2012). In this method, after removing tobacco from CBs, the filter rod in CBs was separated and used in IFAS as an attached growth bed.

Production of superhydrophobic/superoleophilic fibers from CB

A method to control CB pollution is the conversion of its filter into useful materials. Hence, by soaking a cigarette filter in NaOH solution and then in an ethanolic solution of hexadecyltrimethoxy silane, the cellulose acetate used in the cigarette filter will be converted either to superhydrophobic/superoleophilic fibers (Ou et al. 2016). Ou et al. (2016) used fibers to remove kerosene from water; it was found that these fibers have over 96% efficiency to absorb kerosene, and this ability was not reduced after 10 cycles. However, the amount of moisture in the collected kerosene in the next cycles was increased; adsorption efficiency was estimated to be in the low level. (Ou et al. 2016).

CB recycling in sound absorbent

Efforts to use CBs as sound absorbent show that acoustic properties of CBs and materials made from cellulose acetate are the same and in some cases they have characteristics better

Table 3 Toxicity level of CBs and nanostructures produced from CBs in different stages of *Anopheles* life

Anopheles life period	CB with remained tobacco		Silver nanostructures made from CB with remained tobacco	
	LC50	LC90	LC50	LC90
First larval stage	66.17	147.40	4.50	9.98
Second larval stage	75.13	153.09	6.28	12.08
Pupal stage	117.99	209.48	8.07	16.49

Table 4 Effect of CBs and nanostructures produced from CBs at the rate of egg hatchability

Substance type added	Concentration (%)	Egg hatchability (%)
CBs with remained tobacco	100	56.6
	200	32.8
	300	19.4
	400	12.4
	500	0
Silver nanostructures made from CBs with remained tobacco	3.125	71.4
	6.25	59.6
	12.5	40.4
	25	0

than commercial sound absorbents (Gómez Escobar and Maderuelo-Sanz 2017; Maderuelo-Sanz et al. 2018). Of course, when smoking occurred, the density, mass, and diameter of cellulose acetate fibers are changed, which is effective in acoustical properties (Maderuelo-Sanz et al. 2018). Also, the absorption ratio depends upon CBs porosity; adsorption efficiency increased when porosity of CBs is increased (Gómez Escobar and Maderuelo-Sanz 2017; Maderuelo-Sanz et al. 2018).

Paper production from CBs

Given that an important part of the cigarette filter is composed of cellulose acetate, a method for CB recycling is using it as cellulose pulp for paper production. In the University of Brasilia, a method for cellulose pulp production was developed by the CB pulping method for 3 h at boiling point in the presence of 1% NaOH (Betânia et al. 2016). The paper made of CB pulp had a weight of 71.14 g/m^2 and 0.24-mm thickness. In addition, these papers had 54.16% whiteness and paper stretching and tensile strength were 1.91%, and 0.8 KN/m^2, respectively. These characteristics indicate that this process can be considered as a suitable method for recycling CBs in paper production, although a low whiteness indicates a lack of adequate purification in terms of residue reduction (Betânia et al. 2016).

Use as metal corrosion inhibitors

Another method to recycle CBs is applying it in order to trap chemical compounds. CBs contain a high amount of nicotine which exhibits anti-corrosion properties, and therefore, it can be used to prevent the corrosion of metal surfaces such as iron (Lucatero et al. 2016). The results of a study published in 2016 showed that water extracted from CBs (WECB) had a significant effect on corrosion inhibition by formation of a surface layer on iron (Lucatero et al. 2016).

Applying as lithium-ion battery membrane separator

The cellulose acetate in the cigarette filters can be separated and accordingly applied for a variety of uses. An application for this purpose is creation of a cellulose-based membrane separator for a high-performance lithium-ion battery. The results of a study on smoked cigarette filter showed that by separating the cellulose acetate from the cigarette filter and removing trapped contaminants by a special process, a nanofiber membrane separator with high performance and safe for lithium-ion batteries is produced (Huang et al. 2015). Also, in another method that is shown in Fig. 2, an asymmetric supercapacitor with high electrochemical performance was assembled (Wang et al. 2016).

CB littering control

Due to the high amount and distribution of CBs (Novotny et al. 1999; Kadir and Mohajerani 2011; Dieng et al. 2014; Yang et al. 2017), as well as the low efficiency of collecting mechanical devices and the possibility of these devices damaging the birds' nest and turtle in places such as the beaches

Table 5 Use of CBs to produce carbon in researches

Carbon type	Production method	Surface area (g/^{2}m)	Qualitative characteristics	Modify or activate with	Reference
Activated carbon	Heat in vacuum medium for 3 h at 230 °C. Then, pour the carbon to a size of 100 μm	117.2	The produced porous activated carbon had a diameter of 50 μm and a length of 10 μm, with numerous microporous structures on the activated carbon	KOH	(Yang et al. 2017)
Charred carbon	Direct pyrolysis with 900 °C for 3 h and in the presence of a constant nitrogen gas equal to 100 cm^3/min and a heating rate of 5 °C/min	597	Activated carbon has a micropore surface of 453 m2/g and a mesoporous surface of 144 m2/g. This charred carbon has a pore diameter of 3.32 n	HNO3-	(Masoudi Soltani et al. 2015)
Porous carbon	Single-stage carbonization using pyrolysis heat equal to 900 °C for 1 h with a heating rate of 5 °C/min	637	–	–	(Yazdi et al. 2012)



Fig. 2 Recycling of CB for supercapacitor with high electrochemical performance production (Wang et al. 2016)

(Ariza and Leatherman 2012; Sarani and Kadir 2013; Mohajerani et al. 2016) and considering CB pollutants and its toxic effect (Kadir and Mohajerani 2011; Ariza and Leatherman 2012; Yazdi et al. 2012; Dieng et al. 2013; Sarani and Kadir 2013), finding methods for CB littering control should be seriously pursued across the world. Despite the fact that CB littering bans can be useful (Novotny et al. 1999; Ariza and Leatherman 2012), various studies tried to control CB litters. The most important methods that received much attention from researchers to control CB litters are listed as follows:

a. Labeling

Writing a warning about the non-biodegradability and toxicity of CB on its packet and also the legal provisions can be effective in CB litters reduction (Novotny et al. 1999).

b. Deposit/return

This method is commonly used for drinking bottles; therefore, it can also be used for cigarettes in such a way that the extra cost is given back when CBs are delivered after cigarettes smoking (Novotny et al. 2009).

c. Waste tax

The enacting legislation taxes on waste materials as an additional cost to cigarettes will both increase prices and reduce consumption; therefore, this method can be applied to dispose CBs properly (Novotny et al. 1999; Novotny et al. 2009).

d. Fines

Creating fines for CB littering and spending these funds on cigarette butt collection (Novotny et al. 2009; Ariza and Leatherman 2012).

e. Manufacturer's responsibility

The pressure on factories to make a finance and take practical action against CBs (Smith and Novotny 2011) is proposed; CBs control can be applied by methods such as extended producer responsibility (EPR) as a postconsumer hazardous waste process. The EPE also used for tire and acid-lead batteries, fluorescent lamps, paint, and insecticides in some areas can be used for CBs. This process will give manufacturers responsibility for CB collection and disposal costs (Ariza and Leatherman 2012).

f. Modification of the cigarette filters

Attempts to produce biodegradable cigarette filters (Novotny et al. 1999; Novotny et al. 2009; Ariza and Leatherman 2012).

g. Education

Extending public education and aspiring feeling responsibility in consumers (Novotny et al. 1999; Novotny et al. 2009; Ariza and Leatherman 2012) is a very effective method in CB littering reduction. For instance, a study focused on the smokers' behavior and attitude revealed that 74% of smokers often disposed CBs as litters (Rath et al. 2012). some studies estimated that the littered CBs ratio is 76% (Patel et al. 2013) and 84% (Lee and Lee 2015) of the smoking disposal way, which indicates the emphasize the effect of the Smoker's behavior on this litter. Researchers also placed the emphasizes on the required equipment to increase access to the ashtray in public places (see Fig. 3) (Widmer and Reis 2010; Smith and Novotny 2011; Ariza and Leatherman 2012). In a study containing 168 ashtrays, it was revealed that CBs placed in ashtrays was about 50%; many ashtrays were only filled by CBs. The number of needed ashtrays was calculated by calculating the number of CBs thrown in ashtrays and the estimated CB number in the coast (Widmer and Reis 2010).

Discussion

CB littering control

A cigarette filter is designed to trap chemicals and toxic materials existing in cigarette smoke in order to reduce its damage for smokers; although it has a positive impact on the



Fig. 3 Portable beach ashtray (PBA). A pen was added on the left of the photograph for scale interpretation (Widmer and Reis 2010)

smokers' health, there is a serious environment concern due to filtered cigarettes consumption. In addition, given that most of CBs are littered by smokers, CBs durability, and the dispersion of this debris in urban and public places and even marine environments, introducing solutions for controlling and recycling these hazardous wastes with high dispersion seriously is required. Regarding CB littering control, the proposed strategies with the purpose of offsetting costs can be categorized into three distinct groups:

a. Establishment of CB littering fines and waste tax for this debris
b. Creating a responsibility for the cigarette producers and consumers
c. Generic education and awareness of the disadvantages of CBs

It seems that considering the great influence of smokers' behaviors on the littered CB quantity, focusing on general education and the smokers' behavior as a long-term plan can be considered as an appropriate solution for CB litters controlling. Also, enactment of legislation for CB littering, at least in some areas such as beaches and public places, can be effective as a short-term plan to the littered CB control. However, given CB management cost, it seems that the waste tax legislation on filtered cigarette and the use of manufacturers' responsibility method would help to manage CB as one of the most important litters.

CB recycling methods

The efforts made by the researchers for CB recycling were based on physical characteristics and chemical properties of CBs. CB recycling methods such as applying them in brick production, asphalt production, vector control, activated carbon and absorbent production, sound absorbing materials, paper production, and as biofilm carrier in wastewater treatment are tested by researchers. In some applications such as bricks or asphalt production, CB processing is not needed and the cigarette can be used in the original shape. However, for other applications, CB is required to be recycled through chemical extraction, pyrolysis, and filter rod extraction. Of course, in most research, disinfection and drying of CBs were carried out before uses. CB recycling methods and their processing are listed in Table 6.

The results of studies show that recycling CBs in different methods can lead to generation of practical products. These studies shown that CBs due to the physical and chemical characteristics have a variety of recyclable species. However, the use of CBs in some recycling methods can have undesirable effects on the final product features, which are listed in Table 7.

Information presented in Tables 6 and 7 show that CB recycling is possible due to its structural characteristics or chemicals trapped in its filter. The use of chemicals trapped in CBs and the use of cellulose acetate are the most important targets for CB recycling, while the use of whole CBs in the recycling process without separating its components in a few applications such as building material production was tested. The chemicals trapped in the CB could be extracted with distilled water and chemical solvents. However, applying these chemicals with the aims of vector and metal corrosion controls requires some consideration in terms of public health and the environment (Mondal et al. 2015) (Lucareto et al. 2016). Cigarette filters made up of cellulose acetate are employed in supercapacitor productions after removing chemicals and without any physical deformation. Of note, cellulose acetate was chemically converted in production of pulp and adsorbents (Yazdi et al. 2012), which subsequently leads to emission of polluted gases and wastewater (d'Heni Teixeira et al. 2017). An important aspect to CB recycling in creation building materials such as bricks and asphalt is the quality of the final product. Addition of CBs to raw material affects the characteristics of the final products including strength, thermal conductivity, density, and porosity (Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Aeslina and Mohajerani 2012; Kadir et al. 2014; Mohajerani et al. 2016; Murugan et al. 2018). The variation of some characteristics is favorable (reduction of brick density); however, decreases in the bricks' strength are considered as unfavorable characteristics. Furthermore, the recycling process in proper operation conditions such as suitable mixing time leads to produce high-quality products.

Table 6 Methods of recycling and reuse of CBs

CB recycling methods	Processing used	References
Vector control	Soak CB in 100 ml distilled water for 24 h and use the chemicals that leak from the CB	(Dieng et al. 2013)
	Soak the CB in tap water at times of 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10 days	(Dieng et al. 2011)
	Soak CB in 100 ml distilled water for 24 h and use the chemicals that leak from the CB	(Dieng et al. 2014)
	Soak 5 g of CB in 250 ml of distilled water for 24 h, then centrifuge the solution and use the supernatant	(Mondal et al. 2015)
	Use the extract from the CB for the construction of nano silver structures in the presence of the AgNo3 solution	(Murugan et al. 2018)
Asphalt production	Encapsulated in bitumen and in paraffin	(Mohajerani et al. 2017)
Clay brick production	No processing	(Kadir and Mohajerani 2011)
	No processing	(Abdul Kadir and Mohajerani 2012)
	No processing	(Abdul Kadir and Mohajerani 2012)
	No processing	(Kadir et al. 2014)
	No processing	(Mohajerani et al. 2016)
	No processing	(Sarani and Kadir 2013)
Activated carbon production	Heat in vacuum medium for 3 h at 230 °C. Then, pour the carbon to a size of 100 μm	
	Direct pyrolysis with 900 °C for 3 h and in the presence of a constant nitrogen gas equal to 100 cm^3/min and a heating rate of 5 °C/min	(Masoudi Soltani et al. 2015)
	Single-stage carbonization using pyrolysis heat equal to 900 °C for 1 h with a heating rate of 5 °C/min	(Yazdi et al. 2012)
Use for biofilm barrier	Extracting the filter rod	(Sabzali et al. 2012)
	Extracting the filter rod	(Sabzali et al. 2011)
Super hydrophobic fibers production	Soak the cigarette filter in a NaOH solution and then in an ethanolic solution of hexadecyltrimethoxysilane	(Ou et al. 2016)
	Soak in hexane solution 4% at room temperature	(Liu et al. 2015)
Porous sound absorber production	Manual removal of cellulose acetate from the cigarette filter	(Murugan et al. 2018)
	No processing	(Maderuelo-Sanz et al. 2018)
Paper pulp production	Pulping the cigarette for 3 h at boiling point and in the presence of 1% NaOH	(d'Heni Teixeira et al. 2017)
Use as corrosion inhibitors for the metal surface	Water extracts of CBs	(Lacureta et al. 2016)
Applying in lithium-ion batteries	The wrapping paper and any remaining tobacco and chars were removed; the cellulose acetate filters were soaked in distilled water, followed by sonication in ethanol with a UP400S ultrasonicator	(Huang et al. 2015)
Supercapacitor production		(Wang et al. 2016)

Although the addition of CBs to raw material due to its adsorption capacity reduces heavy metal leakage in the bricks production process, in all methods for CBs recycling, pollutant emission is occurring in the gas or liquid form. In some cases, researchers regardless of the use of chemical treatment processes could not reduce the pollutant from wastewater due to CBs recycling to an acceptable level (Betânia et al. 2016). Also, in some methods of recycling, a specific part of CBs is separated, other components remain as waste and require management. Fortunately, trapped chemicals released from CBs in recycling processes can be used for some purposes such as vector control and metals corrosion inhibitors (Dieng et al. 2013, 2014; Authors 2016). Overall, the high quality of production derived from CB recycling methods are reported in many research (Dieng et al. 2014). For instance, toxins found in CBs are highly effective in the field of vector control. In addition, activated carbons derived from CBs are of high active surface (600 m^2/g) which are suited for treatment application such as water and wastewaters (Masoudi Soltani et al. 2015) (Yang et al. 2017).

Regarding to the quality and efficiency of the products derived from CBs (see Table 7), we can choose the most suitable methods for CB recycling. Given the changes in qualitative characteristics of bricks following CBs addition and as well as moderate adsorption capacity of carbon derived from

Table 7 CB recycling methods' results

CB recycling methods	Remarks	References
Clay brick production	1. Pollutant emission: The bricks production with the 5% CBs was released CO, CO_2, HCN, and NO 5.8, 2.76, 2.5, and 0.5 times higher than bricks without CB, respectively. The variation in the emissions of these gases in bricks with cigarette butt was reported differently at varying thermal rates 2. Energy saving: Due to the cellulose acetate in CBs that has a thermal value, bricks production with CBs reduces energy consumption. Adding CBs (5%) to the raw material leads to energy savings (58%) in bricks production 3. Properties such as density, thermal conductivity, porosity, strength, and water absorption capacity in the final product will change due to the adding of CBs in the raw material. By adding CBs, the bricks' density and heat conductivity are decreased but the bricks' porosity and water absorption capacity are increased. Cellulose acetate burning in the bricks production process leads to decrease in CBs' density and heat conductivity so lighter bricks produce which transmission and installation costs decrease, also the final product strength decreases 4. The bricks' thermal conductivity depends on their density and porosity, and by adding CBs (5 and 10%) to the bricks material reduced their thermal conductivity rate (51 and 58%), which can be very important in energy-saving issues 5. Factors such as the mixing rate and the heating rate in the bricks production process and also CBs ratio in raw material are effective in final quality. So that increasing the mixing time lead to decreased bricks' density and porosity and increased bricks' strength and water absorption capacity and also bricks' thermal conductivity due to more separating of the cellulose acetate fibers. In lower heating rate, bricks' characteristics are improved, so researchers have found that 2.5% CB and heating rate of 1 °C/min are ideal production conditions	(Kadir and Mohajerani 2011; Abdul Kadir and Mohajerani 2012; Sarani and Kadir 2013; Mohajerani et al. 2016)
Activated carbon production	1. The carbon derived from CBs has a moderate adsorption capacity compared to other adsorbents 2. Lead absorption capacity of charred carbon derived from CBs was improved by KOH (74.5 mg/g compared with 66 mg/g for non-modified charred carbon) 3. Activated carbon derived from the cigarette filter had a high methylene blue absorption capacity (88.77 mg/g)	(Masoudi Soltani et al. 2015; Yang et al. 2017)
Vector control	1. CBs' toxicity level depends on some factors such as the cigarette characteristics, type of plant, and the part of the plant used for cigarette production, and filter characteristics such as porosity and smokers' behavior, which affects the chemical content on CB 2. CB toxicity for larvae is due to toxic substances leaching from it and the remaining tobacco in it 3. CBs toxicity in the mosquitoes' primary life stages is more than the maturity stages, so this condition in the early larval stage is more than the further larval stage, and also in the larval stage over the pupal stage 4. CBs are not only toxic for mosquitoes but also toxic for the next generation of these vectors. It has been observed that the lifespan of adult insects exposed to CBs were reduced, as well as CB effect on the fertility and lifespan in the next generation of *Aedes aegypti*	(Dieng et al. 2011, 2013, 2014; Masoudi Soltani et al. 2015)
Used for biofilm barrier	1. The IFAS system with filter rods derived from the cigarette filter has a good performance in pollutant reduction in wastewater treatment more than the active sludge process 2. The filter rods are 10 mm long and 7 mm in diameter, with 81 to 85% porosity, which occupied 5% of the IFAS reactor volume 3. Using the filter rod in the Anaerobic Moving Bed Biofilm Reactors (AMBBRs) as an attached growth bed determined	(Sabzali et al. 2011, 2012)

Table 7 (continued)

CB recycling methods	Remarks	References
	that the filter rod function had no significant difference with the performance of the Kaldnes-k_3 as a commercial bed	
Paper pulp production	1. The main problem in this process is strong dark liquor generation which has very high COD 2. This dark liquor return to the paper production process from CBs can be effective in reducing its quantity	(d'Heni Teixeira et al. 2017)
Sound absorber production	1. Various samples made in a thickness of 26.5 mm had a high absorption capacity for sound, and with increasing frequency, the absorption coefficient increased, and at above 2000 Hz it reached more than 0.8 in all samples made and at frequencies exceeding 4800 Hz reached more than 0.9. 2. CBs can act as a sound absorber and can compete with other commercially available sound-absorbing products 3. No significant differences between the used cigarette filter and the unused filter in the sound absorption property 4. The sound absorption increase with density and porosity increasing	(Gómez Escobar and Maderuelo-Sanz 2017; Maderuelo-Sanz et al. 2018)
Metal corrosion inhibitors production	1. The free energy data and the Temkin adsorption isotherm process show that the adsorption process of water extracted from CB (WECB) on the metal surface follows a physisorption 2. The WECB-coated metal surface analyzed by scanning electron microscopy confirms the corrosion inhibition of the water extract of CBs in the acid medium 3. The current densities of both cathodic and anodic branches are greatly affected in the presence of WECB in the corrosive medium	(Lucatero et al. 2016)
Supercapacitor production	1. Final product shows good tensile strength (34.1 MPa), high porosity (66%), excellent thermal stability (to 200 °C), and super electrolyte compatibility (355% electrolyte uptake) 2. It has a lower interfacial resistance (98.5 Ω) and higher ionic conductivity (6.16 mS cm^{-1}) than those of commercial separators (280.0 Ω and 0.88 mS cm^{-1}) 3. The rate capability (138 mAh g^{-1}) and cycling performance (75.4% after 100 cycles) are also superior to those of the commercial separators, demonstrating the cellulose-core fibrous membrane to be a promising separator for a high-power and more secure lithium-ion battery	(Huang et al. 2015)

CBs compared to other adsorbents, these methods do not have the potential for CBs recycling in a real scale. Among the proposed methods, superhydrophobic/superoleophilic fiber production can be an attractive alternative to CB recycling due to the quality of the product and no performance loss for several cycles. CB recycling for supercapacitor production can also be considered as an acceptable method for CB recycling, considering the product quality in comparison with its commercial types. Concerns about the use of CB toxins or release of pollutants in the recycling process can be the limitations of some recycling methods such as vector control, metal corrosion inhibitors production, and paper production.

Concerns about the recycling of hazardous wastes such as CBs need more attention in terms of environmental issue in recycling plans. For example, it is mentioned that *Aedes aegypti* has greater tendency to spawn (egg laying) in water containing CBs than without CBs, so the presence of CBs is not a factor in changing the mosquito laying site (Dieng et al. 2014), as well as the chemicals extracted from CBs did not reduce *Anopheles stephensi's* predator activity (Murugan et al. 2018).

The variation in CB recycling methods shown in Table 6 and Table 7 is a potential to integrate several methods to reduce the concern about the hazardous residue in a particular use of CB. For example, the use of cellulose acetate for the sound absorber production or supercapacitor production after extraction of the trapped chemicals for corrosion inhibitors production or carrier control purpose can reduce the wastewater and hazardous waste production.

Conclusion

The methods of control and recycling cigarette butt (CB) as a major litter was systematically reviewed. Regarding the effect of

smokers' behavior on CB littering, attempts to modify their behavior have a significant effect on littered CB reduction. Also, applying taxes and penalty and improving the producer responsibility feelings can improve the management of CBs as a hazardous waste. Studies have shown that many characteristics of CBs, even trapped chemical and toxic materials, can be considered as an opportunity to produce appropriate products from this waste. The CB recycling can be defined in three ways:

I. The use of the whole CBs without any processing method, use of this waste in bricks and asphalt production
II. The use of chemicals trapped in the CBs for purposes such as vector control and corrosion inhibitors
III. The use of cellulose acetate existing in cigarette filter to create suitable materials such as activated carbon, porous sound absorber, and super hydrophobic fibers.

Changes in the final product quality and the pollutant emission are the main problems of the CB recycling methods. However, operation in proper conditions and the integration of various CB recycling methods make it possible to reduce these problems. Nevertheless, environmental aspects of the CB recycling as a hazardous waste need further study.

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References

Abdul Kadir A, Mohajerani A (2012) Properties improvement of fired clay bricks incorporating with cigarette butts. Adv Mater Res 535–537:1723–1730. https://doi.org/10.4028/www.scientific.net/AMR.535-537.1723

Aeslina AK, Mohajerani A (2012) Leachability of heavy metals from fired clay bricks incorporated with cigarette butts. ISBEIA 872–877. https://doi.org/10.1109/ISBEIA.2012.6423017

Ariza E, Leatherman SP (2012) No-smoking policies and their outcomes on U.S. beaches. J Coast Res 278:143–147. https://doi.org/10.2112/JCOASTRES-D-10-00137.1

Authors F (2016) Corrosion inhibition studies of cigarette waste on the iron surface in acid medium : electrochemical and surface. doi: https://doi.org/10.1108/ACMM-05-2014-1384

Barnes RL (2011) Regulating the disposal of cigarette butts as toxic hazardous waste. Tob Control 20:45–48. https://doi.org/10.1136/tc.2010.041301

Betânia M, Antônio M, Duarte B, et al (2016) Process development for cigarette butts recycling into cellulose pulp 2010:. doi: https://doi.org/10.1016/j.wasman.2016.10.013

d'Heni Teixeira MB, Duarte MAB, Raposo Garcez L, Camargo Rubim J, Hofmann Gatti T, Suarez PAZ (2017) Process development for cigarette butts recycling into cellulose pulp. Waste Manag 60:140–150. https://doi.org/10.1016/j.wasman.2016.10.013

Dieng H, Saifur RGM, Ahmad AH, Md Rawi CS, Boots M, Satho T, Zuharah WF, Fadzly N, Althbyani A, Miake F, Jaal Z, Abubakar S (2011) Discarded cigarette butts attract females and kill the progeny of Aedes albopictus. J Am Mosq Control Assoc 27:263–271. https://doi.org/10.2987/11-6124.1

Dieng H, Rajasaygar S, Ahmad AH, Ahmad H, Rawi CSM, Zuharah WF, Satho T, Miake F, Fukumitsu Y, Saad AR, Ghani IA, Vargas REM, Majid AHA, AbuBakar S (2013) Turning cigarette butt waste into an alternative control tool against an insecticide-resistant mosquito vector. Acta Trop 128:584–590. https://doi.org/10.1016/j.actatropica.2013.08.013

Dieng H, Rajasaygar S, Ahmad AH, Md. Rawi CS, Ahmad H, Satho T, Miake F, Zuharah WF, Fukumitsu Y, Saad AR, Abdul Hamid S, Vargas REM, Ab Majid AH, Fadzly N, Abu Kassim NF, Hashim NA, Ghani IA, Abang FB, AbuBakar S (2014) Indirect effects of cigarette butt waste on the dengue vector Aedes aegypti (Diptera: Culicidae). Acta Trop 130:123–130. https://doi.org/10.1016/j.actatropica.2013.11.001

Gómez Escobar V, Maderuelo-Sanz R (2017) Acoustical performance of samples prepared with cigarette butts. Appl Acoust 125:166–172. https://doi.org/10.1016/j.apacoust.2017.05.001

Huang F, Xu Y, Peng B, Su Y, Jiang F, Hsieh YL, Wei Q (2015) Coaxial electrospun cellulose-core fluoropolymer-shell fibrous membrane from recycled cigarette filter as separator for high performance lithium-ion battery. ACS Sustain Chem Eng 3:932–940. https://doi.org/10.1021/acssuschemeng.5b00032

Kadir AA, Mohajerani A (2011) Recycling cigarette butts in lightweight fired clay bricks. P I Civ Eng - Constr Mater 872–877. https://doi.org/10.1109/ISBEIA.2012.6423017

Kadir AA, Sarani NA, Leman AM (2014) Testing on building material using waste material in fired clay brick. Mater Sci Forum 803:330–336. https://doi.org/10.4028/www.scientific.net/MSF.803.330

Lee W, Lee CC (2015) Developmental toxicity of cigarette butts—an underdeveloped issue. Ecotoxicol Environ Saf 113:362–368. https://doi.org/10.1016/j.ecoenv.2014.12.018

Liu C, Chen B, Yang J, Li C (2015) One-step fabrication of superhydrophobic and superoleophilic cigarette filters for oil-water separation. J Adhes Sci Technol 29(22):2399–2407. https://doi.org/10.1080/01694243.2015.1062653

Lucatero LMB, Ortega DT, Pandiyan T, Singh N, Singh H, Sarao TPS (2016) Corrosion inhibition studies of cigarette waste on the iron surface in acid medium: electrochemical and surface morphology analysis. Anti-Corros Method M 63(4):245–255. https://doi.org/10.1108/ACMM-05-2014-1384

Maderuelo-Sanz R, Gómez Escobar V, Meneses-Rodríguez JM (2018) Potential use of cigarette filters as sound porous absorber. Appl Acoust 129:86–91. https://doi.org/10.1016/j.apacoust.2017.07.011

Masoudi Soltani S, Kazemi Yazdi S, Hosseini S, Bayestie I (2015) Lead removal from aqueous solution using non-modified and nitric acid-modified charred carbon from the pyrolysis of used cigarette filters. Desalin Water Treat 53:126–138. https://doi.org/10.1080/19443994.2013.835751

Mohajerani A, Kadir AA, Larobina L (2016) A practical proposal for solving the world's cigarette butt problem: recycling in fired clay bricks. Waste Manag 52:228–244. https://doi.org/10.1016/j.wasman.2016.03.012

Mohajerani A, Tanriverdi Y, Nguyen BT, Wong KK, Dissanayake HN, Johnson L, Whitfield D, Thomson G, Alqattan E, Rezaei A (2017) Physico-mechanical properties of asphalt concrete incorporated with encapsulated cigarette butts. Constr Build Mater 153:69–80. https://doi.org/10.1016/j.conbuildmat.2017.07.091

Mondal NK, Hajra A, Chakraborty D, et al (2015) Full length research paper cigarette butt waste and its effective utilization towards larvicidal activity of mosquito 3:9–15

Murugan K, Suresh U, Panneerselvam C, Rajaganesh R, Roni M, Aziz AT, Hwang JS, Sathishkumar K, Rajasekar A, Kumar S, Alarfaj AA, Higuchi A, Benelli G (2018) Managing wastes as green resources: cigarette butt-synthesized pesticides are highly toxic to malaria

vectors with little impact on predatory copepods. Environ Sci Pollut Res 25:10456–10470. https://doi.org/10.1007/s11356-017-0074-3

Novotny TE, Zhao F, Novotny TE, Zhao F (1999) Consumption and production waste: another externality of tobacco use. Tob Control 8:75–80

Novotny TE, Lum K, Smith E, Wang V, Barnes R (2009) Cigarettes butts and the case for an environmental policy on hazardous cigarette waste. Int J Environ Res Public Health 6:1691–1705. https://doi.org/10.3390/ijerph6051691

Ou J, Wan B, Wang F, Xue M, Wu H, Li W (2016) Superhydrophobic fibers from cigarette filters for oil spill cleanup. RSC Adv 6:44469–44474. https://doi.org/10.1039/c6ra01303a

Patel V, Thomson GW, Wilson N (2013) Cigarette butt littering in city streets: a new methodology for studying and results. Tob Control 22:59–62. https://doi.org/10.1136/tobaccocontrol-2012-050529

Rath JM, Rubenstein RA, Curry LE, Shank SE, Cartwright JC (2012) Cigarette litter: Smokers attitudes and behaviors. Int J Environ Res Public Health 9:2189–2203. https://doi.org/10.3390/ijerph9062189

Sabzali A, Nikaeen M, Bina B (2011) Evaluation of cigarette filters rods as a biofilm carrier in integrated fixed film activated sludge process 5:101–106

Sabzali A, Nikaeen M, Bina B (2012) Performance evaluation of cigarette filter rods as a biofilm carrier in an anaerobic moving bed biofilm reactor. Environ Technol 33:1803–1810. https://doi.org/10.1080/09593330.2011.646317

Sarani NA, Kadir AA (2013) Thermal conductivity of fired clay bricks incorporated with cigarette butts. Adv Mater Res 690–693:919–924. https://doi.org/10.4028/www.scientific.net/AMR.690-693.919

Smith EA, Novotny TE (2011) Whose butt is it? Tobacco industry research about smokers and cigarette butt waste. Tob Control 20:2–9. https://doi.org/10.1136/tc.2010.040105

Suárez M, Macías C (2017) An experimental demonstration that house finches add cigarette butts in response to ectoparasites. J Avian Biol 48(10):1316–1321. https://doi.org/10.1111/jav.01324

Wang Y, Jiang M, Yang Y, Ran F (2016) Hybrid electrode material of vanadium nitride and carbon fiber with cigarette butt/metal ions wastes as the precursor for supercapacitors. Electrochim Acta 222:1914–1921. https://doi.org/10.1016/j.electacta.2016.12.003

Widmer WM, Reis RA (2010) An experimental evaluation of the effectiveness of beach ashtrays in preventing marine contamination. Braz Arch Biol Technol 53:1205–1216. https://doi.org/10.1590/S1516-89132010000500018

Yang HM, Zhang DH, Andas J, Rahman MLA (2017) Preparation of creating active carbon from cigarette filter waste using microwave-induced KOH activation Preparation of creating active carbon from cigarette filter waste using microwave-induced KOH activation

Yazdi SK, Soltani SM, Hosseini S (2012) An investigation into the optimum carbonization conditions for the production of porous carbon from a solid waste. Adv Mater Res 587:88–92. https://doi.org/10.4028/www.scientific.net/AMR.587.88

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